Customer Churn Prediction & Risk Segmentation Dashboard: Week 2 Summary Report

Project Title: Customer Churn Prediction & Risk Segmentation Dashboard

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Overview :

This project uses Machine Learning to predict telecom customer churn. It analyses customer data to identify users who may leave the service. Logistic Regression achieved the best performance with around 81% accuracy. The project helps businesses improve customer retention and reduce revenue loss.

Dataset Information :

Total records: 7,043 customers

Source : Kaggle-Telecom Customer Dataset

Data Preprocessing Steps:

- Handled missing values in Total Charges
- Created a new feature called Charges Per Month
- Converted categorical data into machine-readable format using **One-Hot Encoding**

Machine Learning Models Compared:

- Logistic Regression
- Random Forest
- XGBoost

Best Performing Model:

- **Logistic Regression**
- Achieved around **81% accuracy**
- Provided better balance and easier interpretation for business use

Main Reasons for Customer Churn:

- Month-to-Month contract customers are more likely to leave
- Customers in their first 6 months have higher churn risk
- High monthly charges increase churn probability

Risk Segmentation:

- 🔴 High Risk (≥ 70%)
    - Immediate customer support and discount offers
- 🟡 Medium Risk (40% – 69%)
    - Marketing campaigns and contract upgrade offers

- 🟢 Low Risk (< 40%)
    - Loyalty rewards and regular updates

Business Recommendations:

- Encourage customers to move from monthly plans to long-term contracts using discounts or free services
- Improve customer support during the first 90 days to increase customer satisfaction and retention

Project Limitations:

- Model uses static customer data only
- Does not include real-time customer behavior like support calls or data usage changes

Future Improvements:

- Use **SMOTE** to handle class imbalance
- Combine Logistic Regression and XGBoost using **Ensemble Stacking** for better performance

Final Outcome:

- The project successfully predicts customer churn and provides useful business insights to improve customer retention strategies.